The Journal of Open Source Software

Psymple: A Python package for complex systems modelling

George Simmons [1¶], **David Stern** [1], **Georg Osang** [1], **Luigi Ponti** [2,3], **Andrew Paul Gutierrez** [3,4], and **Chiara Facciolà** [1]

1 IDEMS International CIC, United Kingdom **2** Agenzia nazionale per le nuove tecnologie, l'energia e lo sviluppo economico sostenibile (ENEA), Italy **3** Center for the Analysis of Sustainable Agricultural Systems (casasglobal.org), USA **4** Division of Ecosystem Science, College of Natural Resources, University of California, Berkeley, USA ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Prashant K Jha

Reviewers:

- @pitmonticone
- @SunnyXu

Submitted: 13 September 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

The modelling of complex systems, which are characterised by high degrees of interdependence amongst components, is continually evolving in response to developing computing and visualisation power. Examples such as ecological, economic or social systems are classically modelled using statistical or correlative techniques, which have natural shortcomings when used for predictive modelling outside of the range of their parameterisation data.

Predictive modelling is improved by deterministic models which capture the evolution of a system, or by hybrid models composed of both deterministic and correlative parts. These models often feature significantly increased size and complexity, and there has been a growing requirement for so-called next generation modelling platforms which are built to a specification requiring modular and data-first implementations which drive clear, adaptable, reusable and accessible modelling practices.

The platform `psymple` is designed to facilitate the development of hybrid complex system models and modelling frameworks. It allows users to link together arbitrary combinations of modular differential equation systems and functional components to build categorical diagrams representing a complex system. A compilation process automatically generates simulatable system equations from these diagrams using the symbolic mathematics package `sympy`. Ultimately, this allows users to focus on the components and interactions of models, rather than their complex equation structure.

Statement of need

The development of `psymple` emerged from the complex system modelling requirements of ecological systems. Ecological niche models, also called species distribution models (SDMs) predict species distributions in response to environmental variables such as geographic and climatic features (Elith & Franklin, 2017; Elith & Leathwick, 2009). Classically, these models are formed using correlative or statistical approaches which match observational data to a set of environmental variables to produce favourability ranges for each species.

An alternative approach is mechanistic modelling, which instead of observational data use physiological data to capture the underlying mechanisms which drive species distribution, such as energy balance, population dynamics or response to climate (Kearney & Porter, 2009). In contrast to correlative approaches, mechanistic SDMs decouple the physiology of a species from their native geography or climate, and allow SDMs in new geographic or climatic regimes to be created in the absence of observational data (Johnston et al., 2019).

An example mechanistic framework is physiologically-based demographic modelling (PBDM), which creates holistic ecosystem models based on the weather-driven biology of component species, allowing for predictive phenology, age- or mass-structured population dynamics, and distribution assessments, see (Gutierrez & Ponti, 2022a, 2022b) for an overview and further references. With this approach, PBDM can account for tritrophic ecosystem interactions (Gutierrez et al., 1999), or model the effects of climate change (Gutierrez et al., 2023).

The schools of thought around correlative SDMs and mechanistic frameworks such as PBDM are seen as largely disjoint (Dormann et al., 2012), but their development can be traced back to early common roots (Fitzpatrick & Nix, 1970; Gutierrez et al., 1974; Wit & Goudrian, 1978). A component of the PBDM framework is the use of physiological data to parametrise "biodemographic" functions capturing biophysical or biochemical mechanisms, such as the development, mortality and fecundity rates of a species in response to environmental variables (Ponti & Gutierrez, 2023).

The use of biodemographic functions in PBDM combines the considerable holistic advantages of mechanistic SDMs, while retaining the comparable ease of parametrisation as status-quo correlative models. More widely, there has been growing interest and development of ecological models explicitly composed of both correlative and mechanistic components (Buckley et al., 2010; Tourinho & Vale, 2023), combining the benefits of both areas. In the wider context of complex systems modelling, the approach of building composite models out of different techniques is called hybrid, or spectrum, modelling.

While PBDM is a highly-developed framework, it shares the same barriers to widespread introduction as general hybrid complex system models. These barriers include the lack of available modelling frameworks (Buckley et al., 2018), the lack of flexibility in existing models (Buckley et al., 2010), or the lack of modelling platforms to implement existing ideas (Ponti & Gutierrez, 2023). The package `psymple` is a general platform designed to facilitate the creation of hybrid complex systems models and modelling frameworks.

Models in `psymple` are built from arbitrary combinations of modular mechanistic, dynamic components and correlative, functional components which naturally interact with each other. This structure allows for the systematic implementation of modelling frameworks such as PBDM, and, more widely, those capturing biological, economic and social systems, for which it is not feasible to capture the laws of interaction purely mechanistically. Examples include agroecological, bioeconomic and Earth systems modelling, and the development of `psymple` is a necessary first step in the development and release of accessible and impactful tools in these areas.

Description

The workings of `psymple` are based on the ideas of the category-theoretic dynamic systems modelling package `AlgebraicJulia.jl` (Libkind et al., 2024), in which modular 'ported' objects, called *resource sharers*, containing ordinary differential equations, expose internal variable states at different ports. Ports are linked by wires which represent the sharing of state variables across the dynamics equations of the system to create complex interacting dynamical systems. Mathematically, resource sharers form algebras over the operad of undirected wiring diagrams, see (Libkind et al., 2022).

In `psymple`, these ideas are extended to realise functions as ported objects, called functional ported objects (FPOs), alongside resource sharers, which in `psymple` are called variable ported objects (VPOs). FPOs contain systems of multivariate functions, whose values are exposed at output ports, and whose arguments can read information in from other FPO outputs or from VPO variable ports. The structure of input ports is also introduced to VPOs by allowing functions defining differential equations to read information in the same way as FPOs.

FPOs and VPOs can be arbitrarily combined into composite ported objects (CPOs) through

the concepts of directed wires, which capture partial substitution of functional information, and variable wires, which implement the functionality of resource sharing. CPOs can themselves read and expose information from input, output and variable ports to create fully modular and arbitrarily complex hybrid systems of both functional and dynamic components whose nested hierarchy reflects the model structure.

More generally, `psymple` is built to a specification shared by "next-generation" dynamical systems modelling frameworks, see (Baez et al., 2023), including being *faceted*, where models can be considered one piece at a time; *modular*, where components naturally compose together; and *functorial*, where the data describing the model (its syntax) is systematically and reliably transformed into system behaviour (its semantics). These ideas allow for legible modelling of highly complex, specialised systems, and drive clear, adaptable and accessible modelling practices in line with those promoted by the Open Modeling Foundation. For example, modular structures pave the way for cross-platform integrations and promote reuse and flexibility, while an abstracted data structure enables the creation of low- or no-code interfaces to improve utilisation amongst non-specialist users.

Concretely, the faceted, modular structure provided by ported objects in `psymple` is complemented by a string-based authoring interface which codifies the diagrammatic wiring structures, and enables no-code authoring from Python dictionary objects or `JSON` file formats. The data-based syntax defining ported objects is internally processed into model semantics, meaning simulatable systems of equations, through a composition process using the Python symbolic mathematics package `sympy` (Meurer et al., 2017). This enables the automatic collection and simplification of equations, the elimination of errors from manually combining complex system equations, and clear inspection of a whole system, or its parts, with automatic outputs including LaTeX format.

Research

The development of `psymple` emerged as part of a collaborative effort between IDEMS International and CASAS Global to increase the uptake, accessibility and impact of the physiologically-based demographic modelling (PBDM) framework. The development of `psymple` is a necessary first step to capture PBDM in sufficient generality to enable modular and flexible implementations of existing and new models into the framework to drive future research and impact potential.

The specification of `psymple` was created from the "collaborative modelling" approach headed-up by the Topos Institute, which uses the mathematical field of applied category theory to capture the mechanisms required to create modelling frameworks collaboratively, accounting for diversity in technical language, data availability and subject expertise. This specification, coupled with the generality and flexibility of `psymple`, enables frameworks beyond PBDM to be developed or implemented, and pushed towards new research and impacts.

For example, research is currently active in bioeconomic modelling, which extends the supply-demand and mass-structured mechanisms of PBDM to understand the economic interaction that humans have with agroecological systems, see (Gutierrez et al., 2020; Regev et al., 1998). More widely, ideas are being conceived around applications in Earth systems modelling, multi-layer modelling incorporating feedback loops to modelling techniques such as collaborative agent-based modelling, and data management and source control across collaborative practices.

Acknowledgements

The collaboration between IDEMS International and CASAS Global is enabled by the McKnight Foundation's Global Collaboration for Resilient Food Systems, grant number $23-149$. The authors would like to thank the members of CASAS Global for their insights in developing the

vision for a modern implementation of PBDM, Tim Hosgood of the Topos Institute for his comments helping put together this paper, and Brendan Fong and the rest of the Topos team in Berkeley for hosting a seminar that kick-started much of the shared vision of this work.

References

Baez, J., Li, X., Libkind, S., Osgood, N. D., & Patterson, E. (2023). Compositional modeling with stock and flow diagrams. *Electronic Proceedings in Theoretical Computer Science*, *380*, 77–96. https://doi.org/10.4204/eptcs.380.5

Buckley, L., Cannistra, A., & John, A. (2018). Leveraging Organismal Biology to Forecast the Effects of Climate Change. *Integrative and Comparative Biology*, *58*(1), 38–51. https://doi.org/10.1093/icb/icy018

Buckley, L., Urban, M., Angilletta, M., Crozier, L., Rissler, L., & Sears, M. (2010). Can mechanism inform species' distribution models? *Ecology Letters*, *13*(8), 1041–1054. https://doi.org/10.1111/j.1461-0248.2010.01479.x

Dormann, C. F., Schymanski, S. J., Cabral, J., Chuine, I., Graham, C., Hartig, F., Kearney, M., Morin, X., Römermann, C., Schröder, B., & Singer, A. (2012). Correlation and process in species distribution models: Bridging a dichotomy. *Journal of Biogeography*, *39*(12), 2119–2131. https://doi.org/10.1111/j.1365-2699.2011.02659.x

Elith, J., & Franklin, J. (2017). *Species distribution modeling*. https://doi.org/10.1016/B978-0-12-809633-8.02390-6

Elith, J., & Leathwick, J. R. (2009). Species distribution models: Ecological explanation and prediction across space and time [Journal Article]. *Annual Review of Ecology, Evolution, and Systematics*, *40*(Volume 40, 2009), 677–697. https://doi.org/10.1146/annurev.ecolsys.110308.120159

Fitzpatrick, E., & Nix, H. (1970). *The climatic factor in Australian grassland ecology*. 3–26. https://api.semanticscholar.org/CorpusID:135291467

Gutierrez, A. P., Nix, H., Havenstein, D., & Moore, P. (1974). The ecology of *Aphis craccivora* koch and subterranean clover stunt virus in south-east australia. III. A regional perspective of the phenology and migration of the cowpea aphid. *Journal of Applied Ecology*, *11*(1), 21–35. http://www.jstor.org/stable/2402002

Gutierrez, A. P., & Ponti, L. (2022a). Analysis of invasive insects: Links to climate change. In *Invasive Species and Global Climate Change* (Second, pp. 50–73). https://doi.org/10.1079/9781800621459.0000

Gutierrez, A. P., & Ponti, L. (2022b). Assessing and managing the impact of climate change on an invasive weed, yellow starthistle. In *Invasive Species and Global Climate Change* (Second, pp. 260–281). https://doi.org/10.1079/9781800621459.0000

Gutierrez, A. P., Ponti, L., Kranthi, K., Baumgärtner, J., Kenmore, P., Gilioli, G., Boggia, A., Cure, J., & Rodríguez, D. (2020). Bio-economics of indian hybrid bt cotton and farmer suicides. *Environmental Sciences Europe*, *32*, 139. https://doi.org/10.1186/s12302-020-00406-6

Gutierrez, A. P., Sabbatini Peverieri, G., Ponti, L., Giovannini, L., Roversi, P., Mele, A., Pozzebon, A., Scaccini, D., & Hoelmer, K. (2023). Tritrophic analysis of the prospective biological control of brown marmorated stink bug, *Halyomorpha halys*, under extant weather and climate change. *Journal of Pest Science*, *96*, 1–22. https://doi.org/10.1007/s10340-023-01610-y

Gutierrez, A. P., Yaninek, J., Neuenschwander, P., & Ellis, C. K. (1999). A physiologically-based tritrophic metapopulation model of the african cassava food web. *Ecological Modelling*,

DRAFT

123(2), 225–242. https://doi.org/10.1016/S0304-3800(99)00144-1

Johnston, A. S. A., Boyd, R. J., Watson, J. W., Paul, A., Evans, L. C., Gardner, E. L., & Boult, V. L. (2019). Predicting population responses to environmental change from individual-level mechanisms: Towards a standardized mechanistic approach. *Proceedings of the Royal Society B: Biological Sciences*, *286*(1913), 20191916. https://doi.org/10.1098/rspb.2019.1916

Kearney, M., & Porter, W. (2009). Mechanistic niche modelling: Combining physiological and spatial data to predict species' ranges. *Ecology Letters*, *12*(4), 334–350. https://doi.org/10.1111/j.1461-0248.2008.01277.x

Libkind, S., Baas, A., Patterson, E., & Fairbanks, J. (2022). Operadic modeling of dynamical systems: Mathematics and computation. *Electronic Proceedings in Theoretical Computer Science*, *372*, 192–206. https://doi.org/10.4204/EPTCS.372.14

Libkind, S., Baas, A., Patterson, E., & Fairbanks, J. (2024). AlgebraicDynamics.jl. In *GitHub repository*. GitHub. https://github.com/AlgebraicJulia/AlgebraicDynamics.jl

Meurer, A., Smith, C. P., Paprocki, M., Čertík, O., Kirpichev, S. B., Rocklin, M., Kumar, A., Ivanov, S., Moore, J. K., Singh, S., Rathnayake, T., Vig, S., Granger, B. E., Muller, R. P., Bonazzi, F., Gupta, H., Vats, S., Johansson, F., Pedregosa, F., … Scopatz, A. (2017). SymPy: Symbolic computing in python. *PeerJ Computer Science*, *3*, e103. https://doi.org/10.7717/peerj-cs.103

Ponti, L., & Gutierrez, A. P. (2023). Challenging the status quo in invasive species assessment using mechanistic physiologically based demographic modeling. *Environment, Development and Sustainability*. https://doi.org/10.1007/s10668-023-03698-9

Regev, U., Gutierrez, A. P., Schreiber, S. J., & Zilberman, D. (1998). Biological and economic foundations of renewable resource exploitation. *Ecological Economics*, *26*(3), 227–242. https://doi.org/10.1016/S0921-8009(97)00103-1

Tourinho, L., & Vale, M. M. (2023). Choosing among correlative, mechanistic, and hybrid models of species' niche and distribution. *Integrative Zoology*, *18*(1), 93–109. https://doi.org/10.1111/1749-4877.12618

Wit, C. de, & Goudrian, J. (1978). *Simulation of ecological processes* (2nd ed.). Centre for Agricultural Publishing; Documentation, Wageningen, The Netherlands. https://doi.org/10.1016/0304-3800(80)90048-4

DRAFT